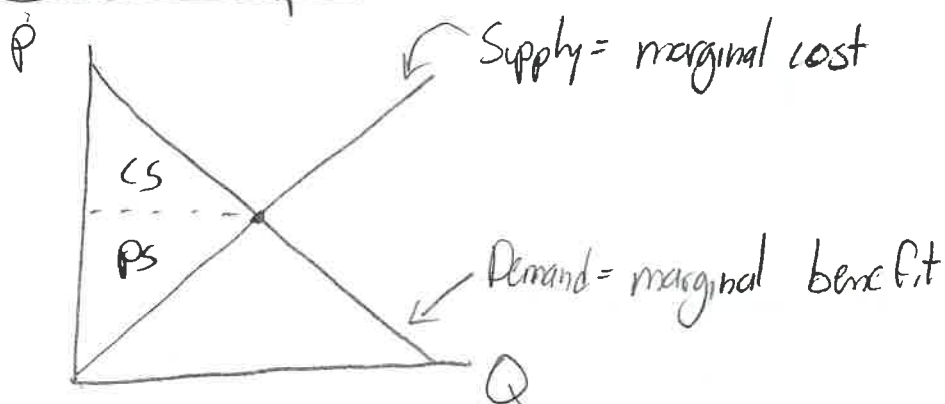
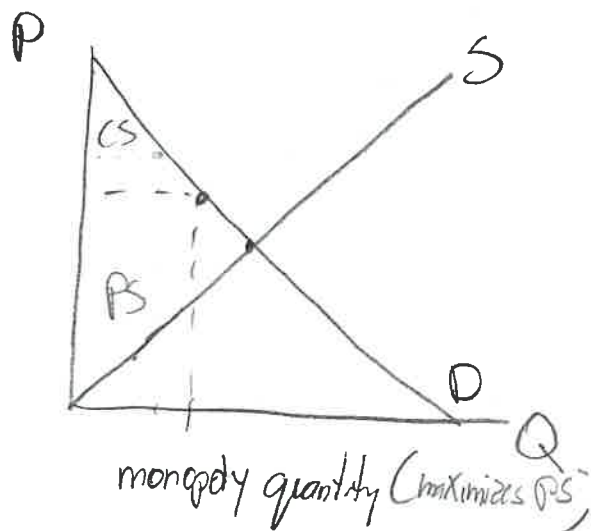


1. Economic Principles



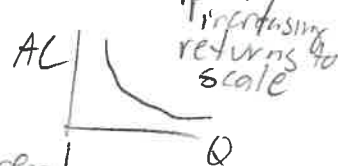
Perfect Competition



Average cost = $\frac{\text{Total Cost}}{\text{Total Output}}$ Marginal cost: cost of next unit (use the supply line)

Natural monopoly: industry where it is most economic to only have one company provide service

Occurs when fixed costs are large relative to average costs



Price discrimination: producer sells the same product to different consumers at different prices based on their willingness to pay.

Can be useful to maximize societal surplus and/or shift surplus between producers and consumers

2. Regulation

Types of regulation:

- (1) Control of entry
- (2) Price fixing
- (3) Prescription of quality
- (4) Imposition or obligation to serve

- Only investor-owned utilities (IOUs) are subject to regulation, not publicly-owned utilities (POUs) or co-ops

Regulatory Compact - regulator grants the company a protected monopoly and the utility agrees to provide non-discriminatory service to all customers within that territory for a fair price

Utility Regulatory Bodies

- Federal Energy Regulatory Commission (FERC)
- State bodies, e.g. public utilities commission

Nominal dollars - what things cost in the year in which they are bought and sold.

$$\text{Real value} = \frac{\text{Nominal value}}{(1 + \text{inflation \%})^{(\text{Nominal \$ year} - \text{Real \$ year})}}$$

$$\text{Nominal discount rate} = (1 + \text{real discount rate})^{(1 + \text{inflation})} - 1$$

4. Revenue Requirement

Capital structure - blend of debt and equity

- Debt ratio: % of total capital that is debt (aka "leverage")
- Equity is higher risk because revenues are first used to pay expenses (including to debtholders) and equity holders only make returns once everyone else has gotten paid

Weighted average cost of capital (WACC) = (Debt ratio * cost of debt) + (Equity ratio * cost of equity)

↳ Measure of the actual cash that is paid to the equity shareholder and bondholders

↑ interest rate

↑ set by the regulator, this is the return on equity

After tax WACC = (Debt ratio * cost of debt * (1 - tax rate)) + (Equity ratio * cost of equity)

↳ Adjust for tax deductions (debt interest is tax-deductible)

Revenue requirement = Operating costs + Depreciation + Return on rate base + Taxes

↳ Employee salaries, rent, fuel, maintenance, PPA contracts

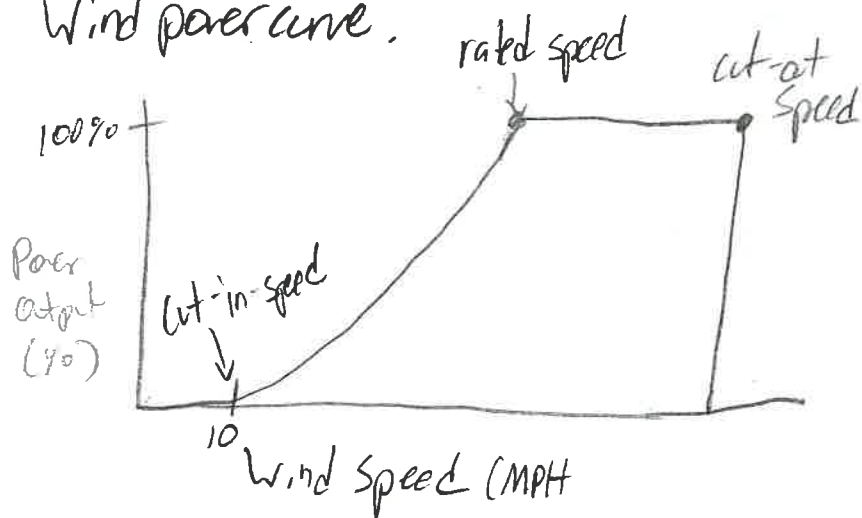
↳ Return "of" initial investment
Straight-line:
• Depreciation rate * capital cost

↳ (Capital cost - accumulative depreciation) * Before tax WACC

↳ rate base from prior year * equity ratio * cost of equity * $\frac{\text{tax rate}}{1 - (\text{tax rate})}$

6. Renewables, Storage, DERs

Wind power curve:

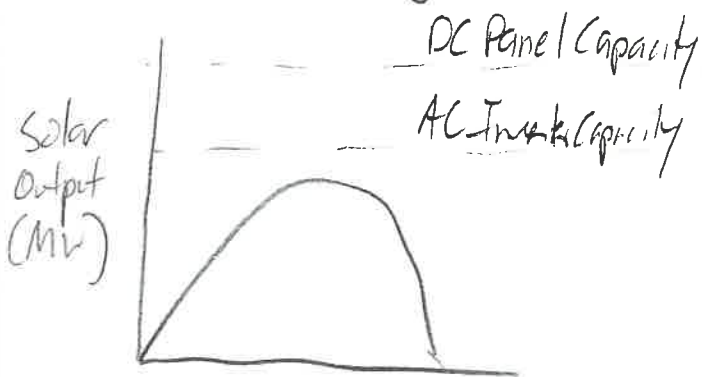


Solar:

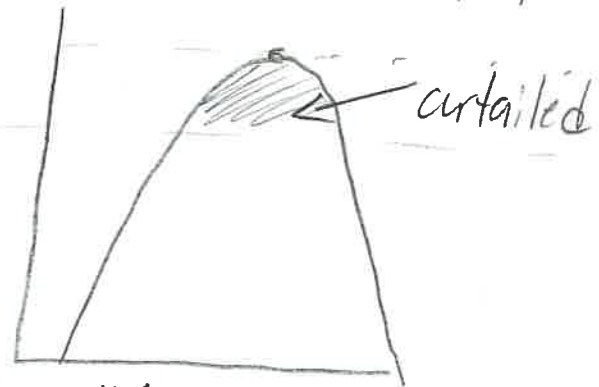
Tilt: how is the panel oriented relative to the ground.

Azimuth: what angle is the panel facing?

Inverter Loading Ratio (ILR): ratio $\frac{\text{DC panel capacity (MW)}}{\text{AC inverter capacity (MW)}}$



Low Solar Gen. Day



High Solar Gen. Day

Solar panels can be fixed or tracking (rotate on 1-2 axes to maximize power output)

Storage:

Power capacity (MW) - max instantaneous charge/discharge

Energy capacity (MWh) - how much energy can be stored?

$$\text{Duration (hr)} = \frac{\text{Energy capacity}}{\text{Power capacity}}$$

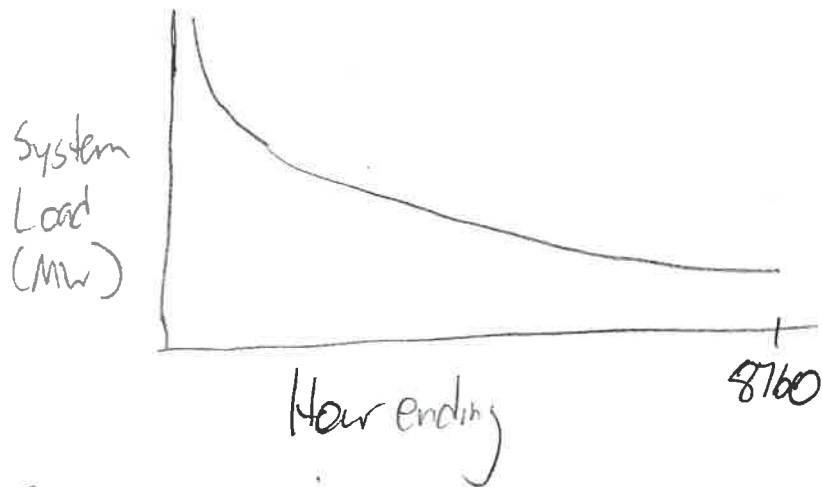
8. System Planning

Goals of system planning:

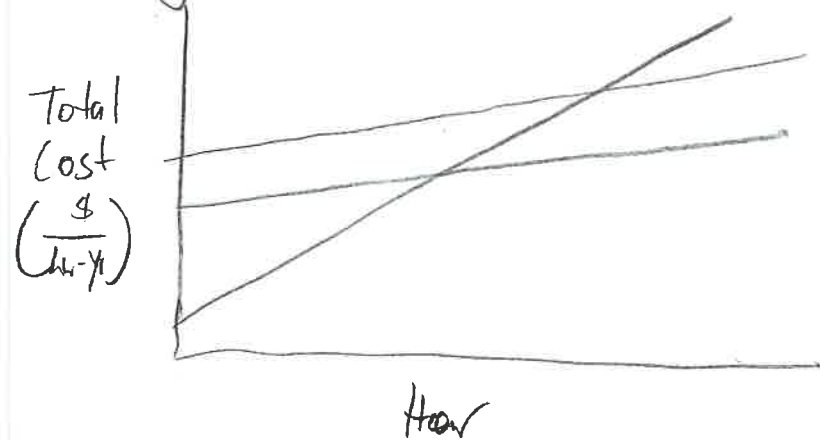
- ① Reliable - enough MW
- ② Operational - enough flexibility to ramp up/down and provide ancillary services
- ③ Environmental - compliant with carbon constraints
- ④ Least-cost - NPV of all fixed + variable costs minimized

Planning reserve margin (PRM): % above forecasted median peak load that helps maintain reliability (~12-20%)

Load duration curve.



Screening curve.



Intercept: fixed cost (\$/kW-yr)
Slope: variable cost (\$/kWh)

Effective Load Carrying Capacity (ELCC) - effective capacity contribution toward the PRM requirement for intermittent (wind, solar) and energy-limited (storage, DR) resources

- ① Tune the system to target reliability standard (1-in-10)
- ② Add ELCC resource to portfolio
- ③ Remove as much perfect capacity as possible without hurting reliability. The amount removed is the ELCC.

10. Marginal Costs

- Which components of the utility's revenue requirements will change due to an increase or decrease in load?

• Energy: dot product $\langle \$/\text{MWh marginal costs}, \text{MWh lightbulb savings} \rangle$

• Energy losses: $\frac{1}{(1 - \text{loss factor})} * (\text{marginal cost of energy})$

• Generation capacity = marginal ELCC. Impose the change, then remove dispatchable generation until we hit the reliability requirement

$$\begin{array}{l} \text{gross} \\ \text{CONE} \end{array} \left[\begin{array}{l} \text{Levelized cost of capacity} \\ + \text{Fixed O\&M} \\ - \text{System Benefits} \end{array} \right] \begin{array}{l} \swarrow \text{adding this resource may cause} \\ \text{more expensive units to be dispatched less} \end{array}$$

Net-CONE $\leftarrow$ net cost of new entry

• Emissions: dot product $\langle \text{ton CO}_2/\text{MWh marginal emissions rate}, \text{MWh lightbulb savings} \rangle$

11. Rate Design

Bonbright Principles - from Principles of Public Utility Rates (1961)

- ① Recovery of total revenue requirement
- ② Fair apportionment of costs among customers

12. Regulatory Cost-Effectiveness

- Cost-effectiveness is a tool for regulators and utilities to assess the impact of various measures on different key stakeholder groups

↳ Participants, ratepayers, utility, state, society

Component	PCT (Participant)	PTM (Ratepayer)	PALT (Utility)	TRC (State)	SLT (Society)
Utility MC		Benefit	Benefit	Benefit	Benefit
Environmental Benefits					Benefit
Uplift/Maintenance Costs	Cost			Cost	Cost
Incentives					
Federal	Benefit			Benefit	Benefit
State	Benefit				
Utility	Benefit	Cost	Cost		
Bill Savings	Benefit	Cost			

13. History of Regulation

1882 - Pearl St. Station in Manhattan - Thomas Edison; i.e. a DC microgrid

1882-1895 - generation/transmission/distribution pattern established, 3-phase AC system (Westinghouse), growth leads to municipal + IOUs, public outcry over too many distribution lines

Early 1900s - state PUCs regulate monopolies, rise of holding companies

1935 - Public Utility Holding Company Act (PUHCA) - holding companies could only be for one integrated system within a single state

1936 - Rural Electrification Act - provided subsidized, low-interest gov't loans to new cooperatively-owned utilities, along with providing training, technical assistance, and equipment, federal income tax exemption

- Public power tends to have lower rates than IOU's because they can buy from federal hydro at cost

North American Grids:

- Western Interconnection
 - Eastern Interconnection
 - ERCOT (TX)
 - Quebec Interconnection
- } Each grid operating synchronously; some DC ties between grids

Balancing Authorities:

- Control power plants + transmission to keep load = gen and comply with NERC standards

Independent System Operators (ISOs):

- Utilities band together to form ISOs to run a wholesale electricity market

- All ISOs are BAs, but not all BAs are ISOs (ex: LADWP)

North American Electric Reliability Corporation (NERC): develop standards for power system operation, monitor, enforce compliance, and educate/train

15/16. Electricity Markets - in ISO territory, ISOs dispatch the electricity system via a bid-based auction process regulated by FERC

Day-Ahead:

- ① ISO forecasts load for the next day
- ② Generators submit bids
- ③ ISO schedules lowest cost generators

- In a 100% renewable grid, 2 regimes are scarcity pricing or capacity pricing

Power Purchase Agreements (PPAs) - out-of-market contract between utility and IPP

Congestion in the transmission grid can lead to different locational marginal prices

17. Deregulation and Customer Choice

Elements of deregulation:

- (1) Sale of generation assets (mostly)
- (2) Establishment of ISOs (always)
- (3) Customer/retail choice (depends)

Pros of Deregulation:

- Eliminate rate base inflation
- Only efficient plants survive

Cons of Deregulation:

- Competitive plants can profit > recovered costs
- Exercise of market power
- Less central planning for an "optimal" resource portfolio

Stranded Costs/Benefits

- During utility's sale of its power plants:

- Asset value < net book $\Rightarrow$ stranded cost (could be recovered through)
- Asset value > net book $\Rightarrow$ Stranded benefit (customer charges)

Retail Choice

- Full: individuals can procure their own generation (TX, NY)
- Partial: community choice aggregators (entire cities) can break away from IOUs and procure their own generation $\rightarrow$ must pay